

OLIST Delivery Performance & Customer Dissatisfaction Analysis

Interim Stakeholder Update

Objective:

Assess whether delivery delays are associated with customer dissatisfaction and identify the main operational areas that require further investigation.

Scope:

Delivered orders linked to valid customer reviews.

Delay analysis based on actual delivery date versus estimated delivery date.

Negative reviews defined as $\text{review_score} \leq 2$.

Method:

Order-level analysis using PostgreSQL outputs imported into Power BI.

The analysis identifies associations, not confirmed root causes.

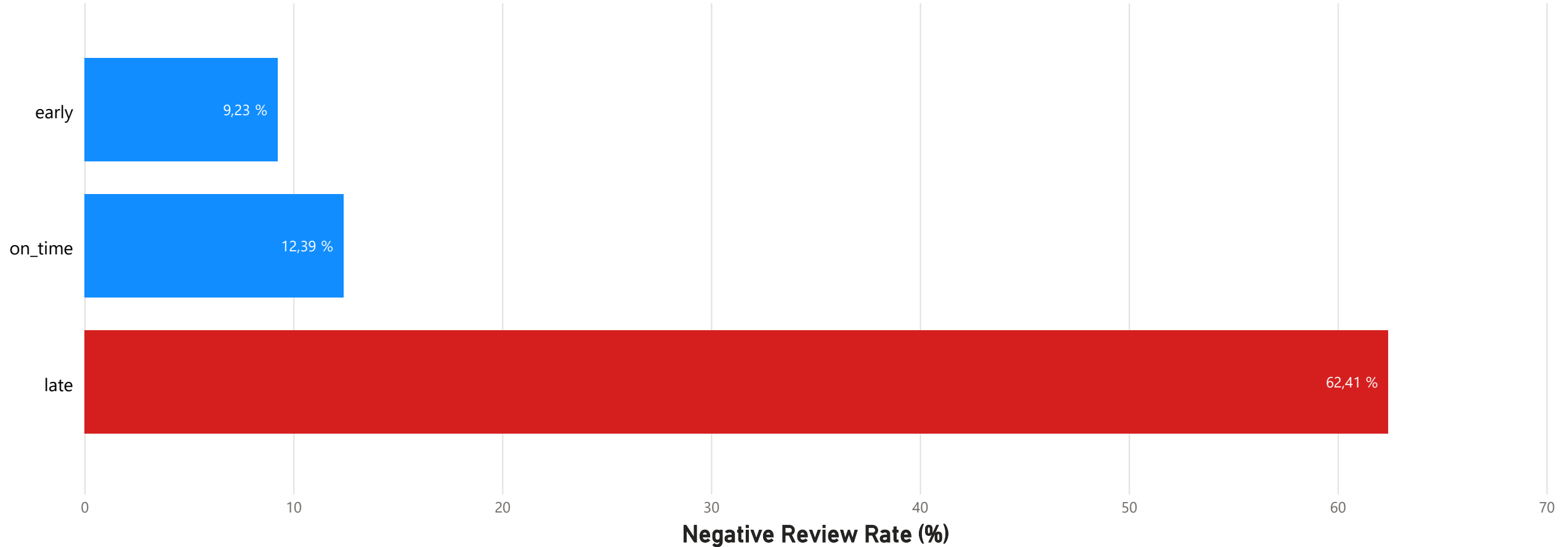
Dataset: OLIST Brazilian E-Commerce

Tools: PostgreSQL, DBeaver, Power BI

Status: Interim analysis — ready for stakeholder review

Small Delay Segment, High Dissatisfaction Impact

Late deliveries account for only 6.65% of reviewed delivered orders, yet 62.41% of them receive negative reviews.



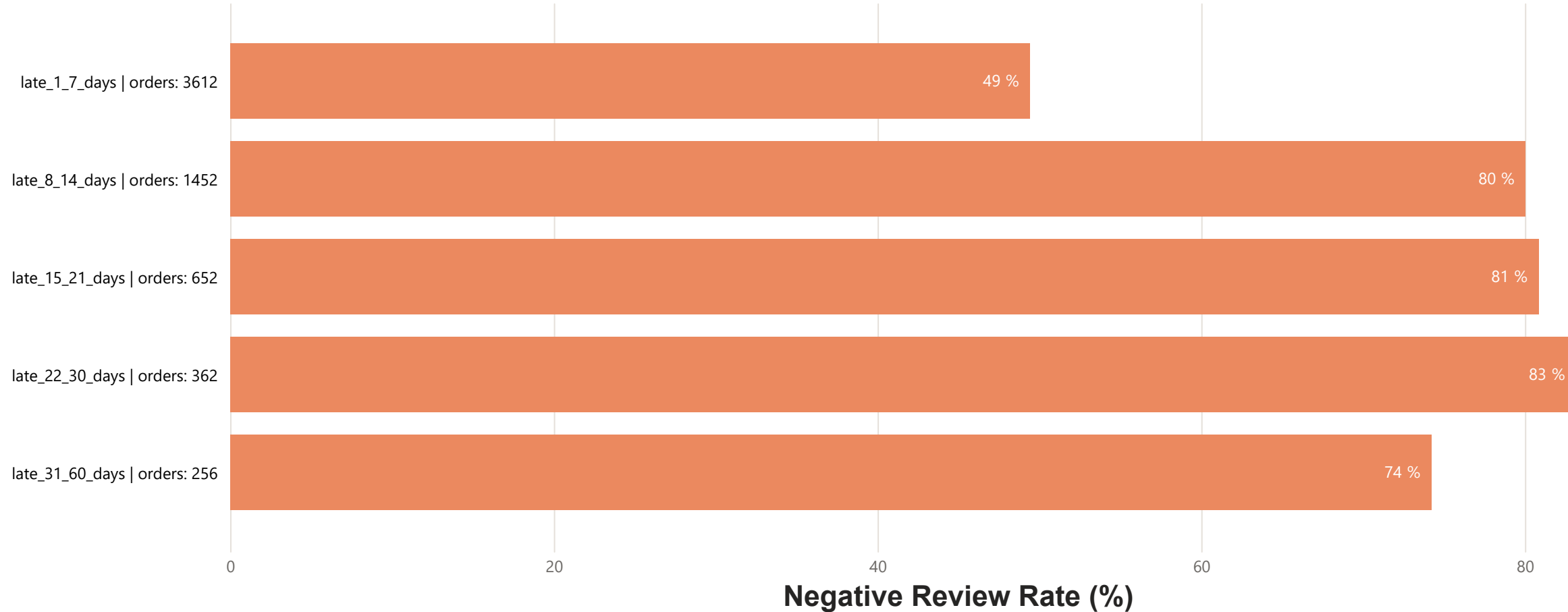
Insight:

Late deliveries represent only **6.65%** of reviewed delivered orders, but they generate a negative review rate of **62.41%**.

This suggests that delivery delay is not a widespread issue, but it has a strong impact when it occurs.

Negative Review Rate by Delay Severity

The 1–7 day delay bucket has the highest volume, while longer delays show stronger dissatisfaction severity.

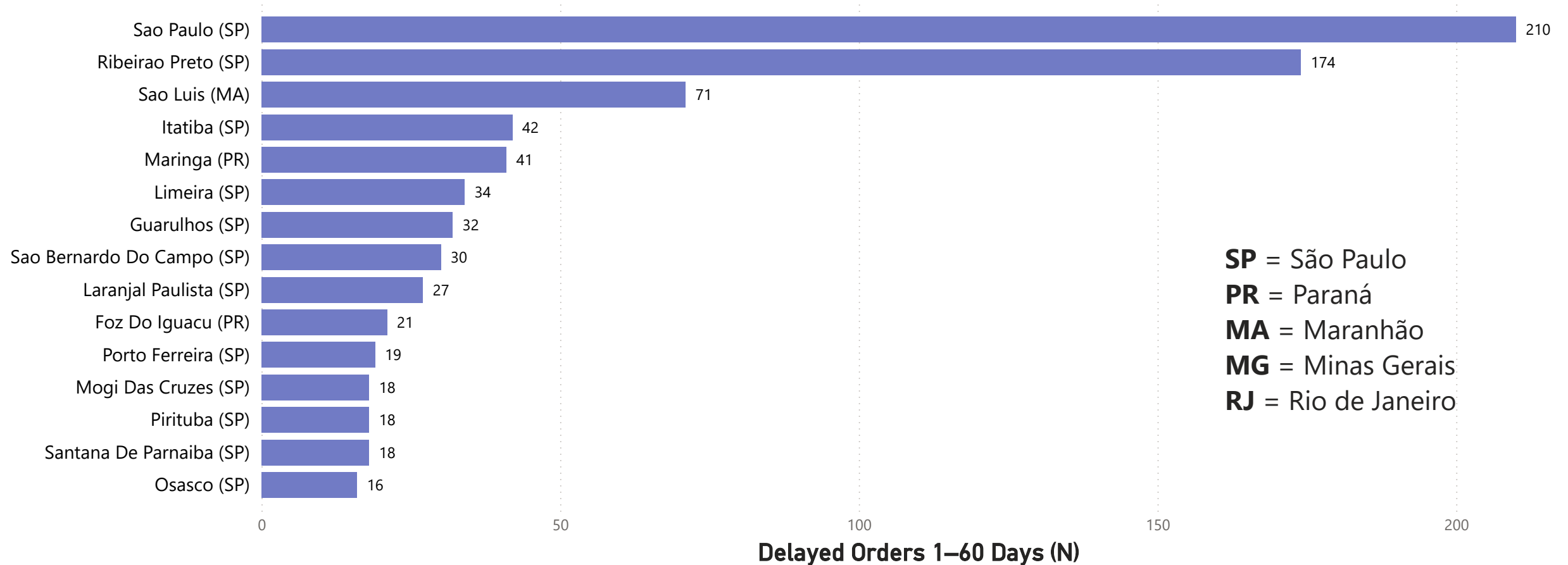


Insight:

The **1–7** day delay bucket has the largest volume with **3,612** orders and a **49.42%** negative review rate. Longer delays show higher severity, with negative review rates around **74%–83%**, but lower order volume.

Critical Delayed Orders by Seller City - Top 15

Critical delayed orders are concentrated in specific seller origin states, especially São Paulo.



Insight:

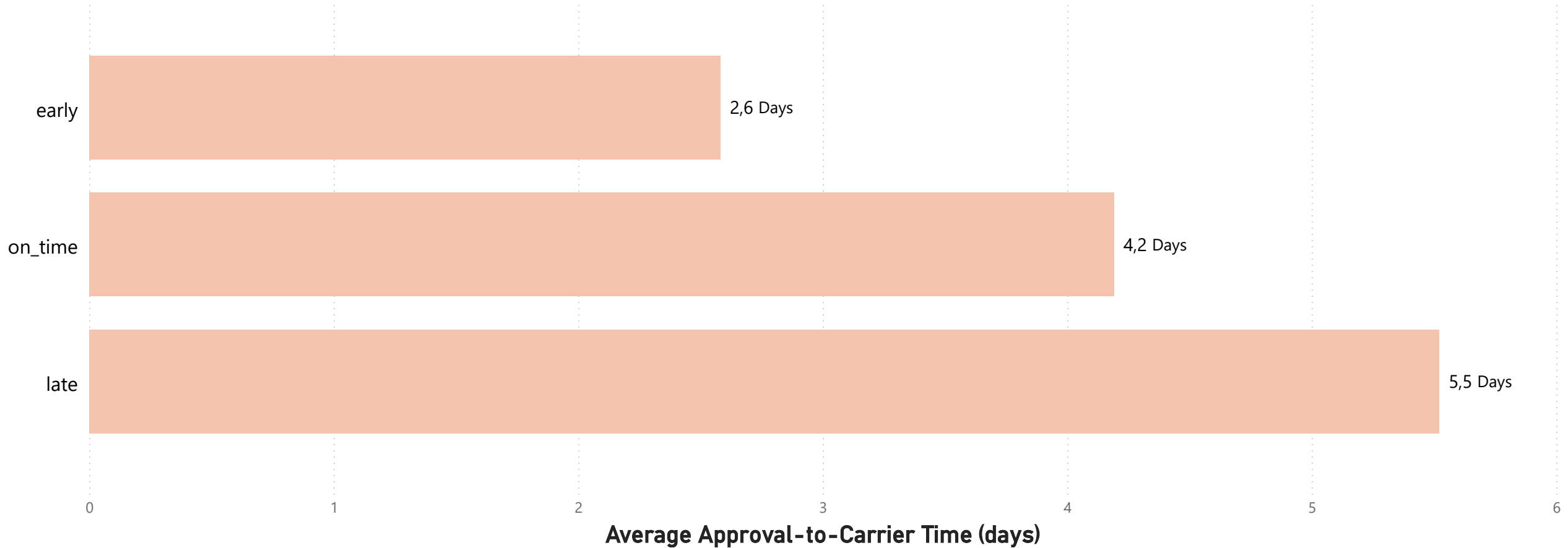
São Paulo (SP) dominates the critical seller-origin areas, with **765** delayed orders in the 1–60 day range.

The next states are far smaller: **PR** = 83, **MA** = 71, **MG** = 38, **RJ** = 34.

This suggests that the issue is concentrated on the seller-origin side, especially in São Paulo.

Average Approval-to-Carrier Time by Delivery Status

Late deliveries show a longer pre-carrier handoff time, suggesting a possible operational issue before final delivery.



Insight:

Late deliveries show the **longest** approval-to-carrier **handoff time**, with an **average** of **5.52** days.

This suggests that part of the delay may start before the final carrier delivery phase, during order preparation, seller processing, warehouse handling, pickup scheduling, or carrier handoff.

Data Needed to Confirm Root Cause

The current analysis identifies strong associations, but additional operational data is required to validate the root cause.

Operational Data Needed

1. Seller/store mapping for critical ZIPs and seller cities.
2. Operational cluster, hub, area manager, logistics partner, or sales channel.
3. Carrier used by order or seller area.
4. Pickup date, dispatch date, and actual carrier handoff date.
5. Seller or warehouse preparation time.
6. Stock availability, order backlog, seller capacity, staffing, and workload indicators.

Business Questions for Stakeholder

1. Which seller_id/store is linked to each critical seller ZIP or city, and what type of location is it: store, warehouse, hub, marketplace partner, or online channel?
2. Are the critical sellers/stores part of internal operational clusters by area, hub, manager, logistics partner, or sales channel?
3. Which carriers were used for orders from the critical seller areas, and are some areas dependent on one carrier or few recurring carriers?
4. What capacity or business constraints affected the critical seller areas: staff, order volume, preparation time, warehouse limits, workload peaks, backlog, stock issues, or known logistics problems?

Decision

Proceed with stakeholder review.

The current data supports a strong association between delivery delays, negative reviews, seller-origin concentration, and longer approval-to-carrier handoff time.

Root cause cannot be confirmed without additional operational data.